

BIOL 202 Large Course Feedback Survey Winter 2025

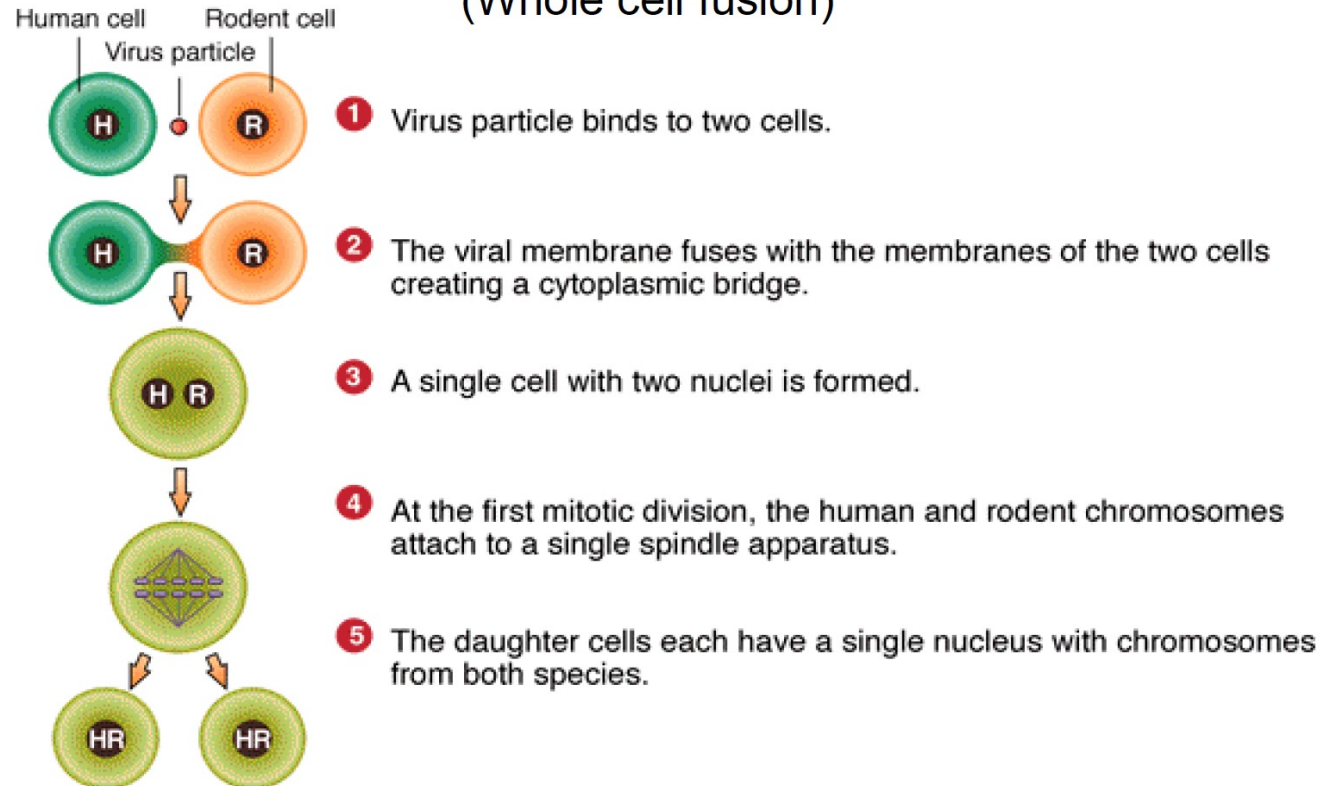


Somatic cell genetics and gene therapy

- Somatic cell fusion
 - ✓ Gene mapping
- Gene therapy
 - ✓ Conventional gene therapy
 - ✓ Genome editing and iPS

Cell hybridization / fusion

(Whole cell fusion)



Cell fusion generates hybrids containing genomes from the two different cell lines

The genomes of fused cells are unstable, randomly keeping or losing chromosomes.

In hybrids between rodent and human cells, rodent chromosomes are predominantly kept and most human chromosomes are lost. (This phenomenon is used to map a GOI in human)

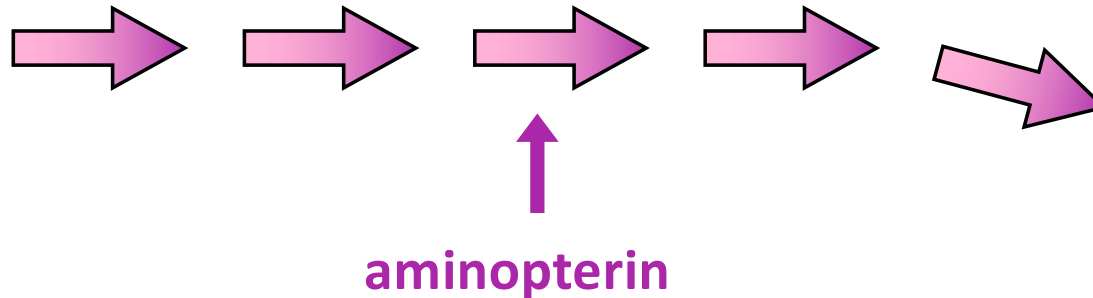
Cell fusion is highly inefficient. Therefore, a selectable marker is necessary to identify hybrids!!!

Natural marker selection : **HAT medium**

hypoxanthine, aminopterin, thymidine

de Novo pathway
(blocked by aminopterin)

Major DNA
precursors



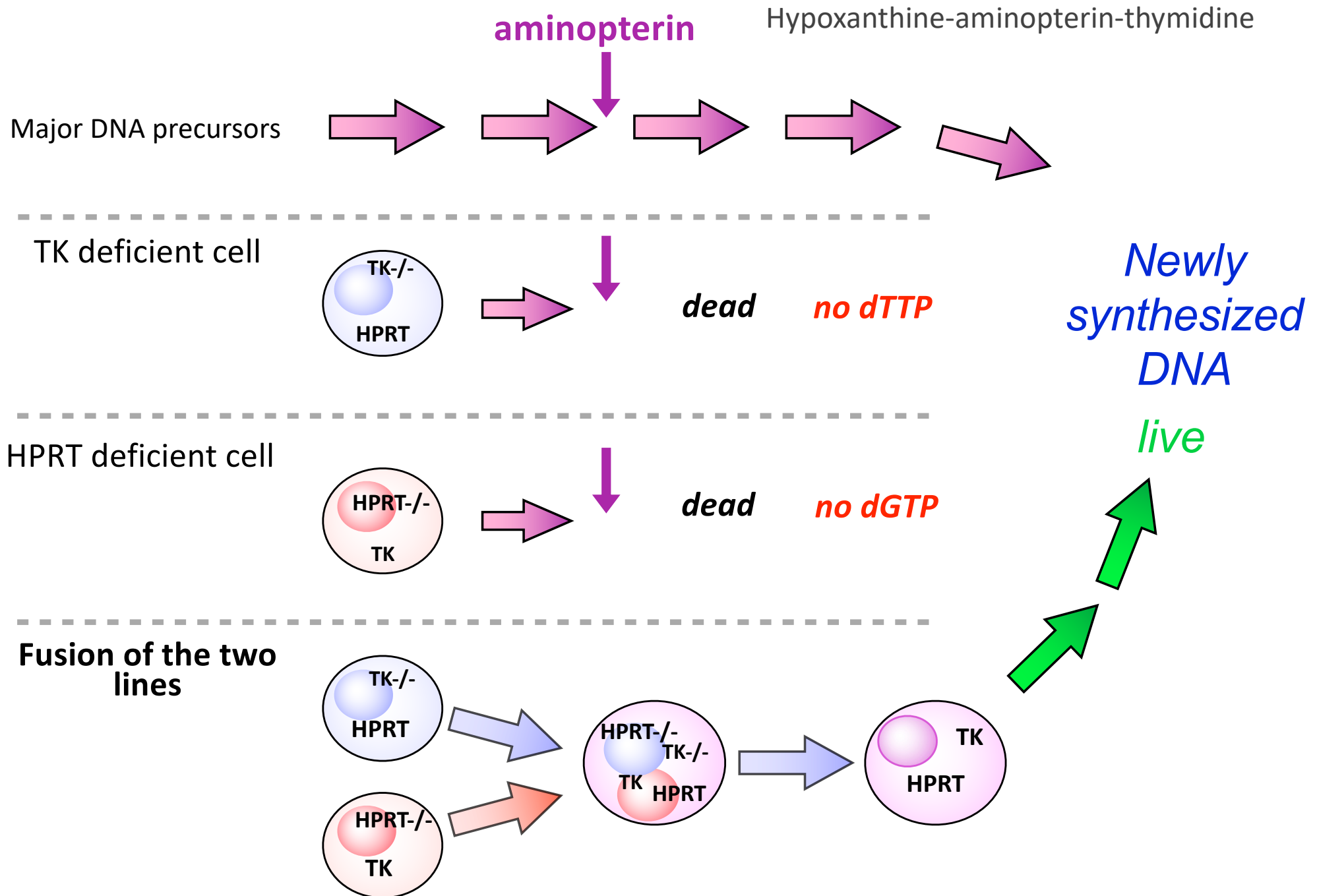
*Newly
synthesized
DNA*

Minor DNA
precursors

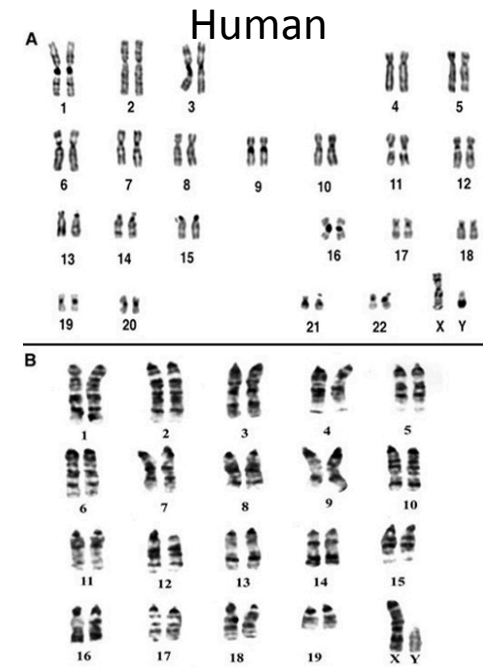
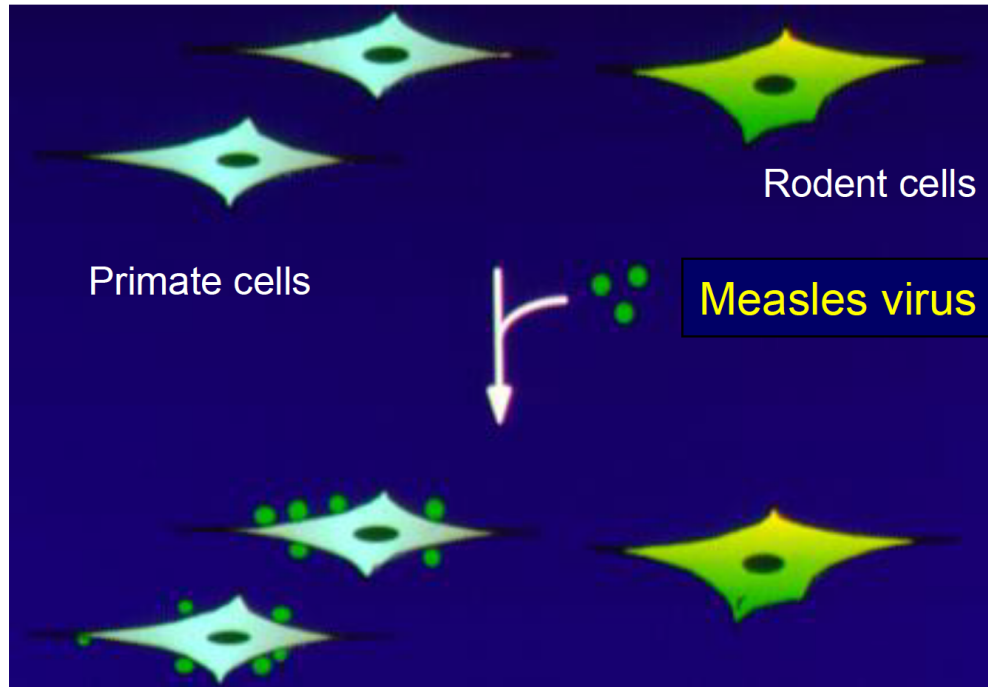


hypoxanthine converted to guanine by Hypoxanthine-guanine phosphoribosyltransferase
thymidine phosphorylated by Thymidine kinase

Natural marker selection : **HAT medium**



Mapping the measles virus receptor in human



Make many hybrids between primate and rodent cells.
(Some human chromosomes will be kept and many will be lost.)

↓

Identify the hybrid cell lines that can be infected by measles.

↓

Determine which human chromosomes are kept in those cells
(structure of human chromosomes are different from rodent chromosomes)

The Human CD46 Molecule Is a Receptor for Measles Virus (Edmonston Strain)

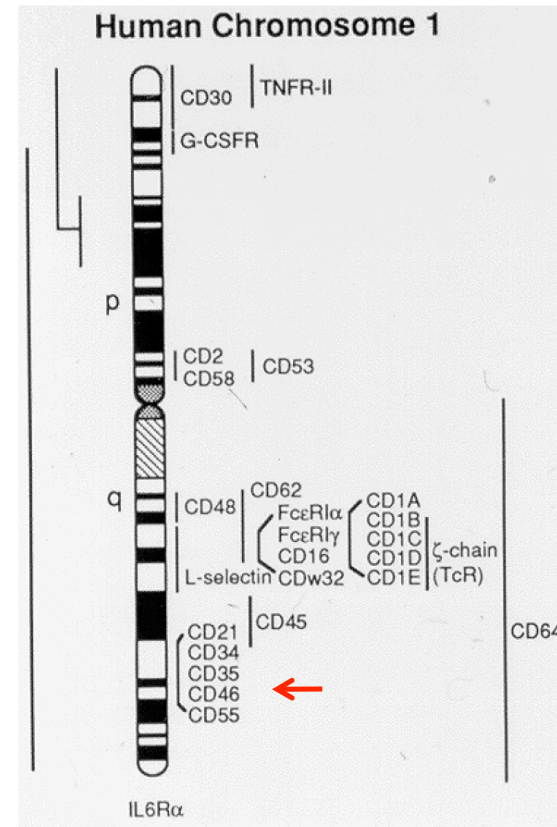
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Human chrom. in hybrid cell line	Virus binding
1, 13, 14	+++
1, X	++++
t(1,17), 14	+++
t(1,21), 2, 18, X	-
3, 5, t(X,21)	-
6, 8, 11, X	-
7, t(X, 11)	-
12, 19	-
21, 22	-



**t(A, B): Translocation between
chromosome A and B**

A cell fusion experiment between human TK-deficient and rodent HPRT-deficient cell lines was carried out to identify the location of the human UMPK gene. Eight different cell lines are recovered from the experiment. The table below shows the presence or absence of the human UMPK activity (+ means present and – means absent) and the human chromosomes retained in each hybrid cell line.

<i>Cell Line</i>	<i>UMPK Activity</i>	<i>Human Chromosomes Present^a</i>
WA-lla	–	2, 3, 4, 11, 12, 14, 16, X
JFA-14b	+	1, 2, 5, 8, 13, 16, 17, 18, X
WA-la	+	1, 2, 4, 7, 8, 10, 12, 13, 15, 16, 18, 19, 20, 21, 22, X
J-10-H-12	+	1, 3, 6, 7, 10, 11, 12, 14, 15, 16, 17, 18, 21, X
AIM-3a	–	2, 4, 5, 7, 8, 10, 11, 12, 13, 14, 17, 18, 20, X
AIM-8a	–	6, 7, 9, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, X
AIM-11a	–	2, 3, 7, 10, 12, X
AIM-23a	+	1, 2, 3, 4, 5, 7, 8, 10, 12, 14, 15, 16, 18, 21, 22, X

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**Which chromosome
encodes the human UMPK
gene?**

① Start presenting to display the poll results on this slide.

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**Which chromosome encodes
the human HPRT gene?**

① Start presenting to display the poll results on this slide.

What is gene therapy and why?

- Gene therapy involves adding a normal (wild-type) copy of a gene to the genome of an individual carrying defective (mutated) copies of the gene, often recessive mutations.
- Only few of the 4000 inherited human diseases are currently treatable.

The viral vectors

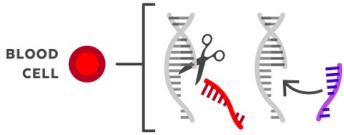
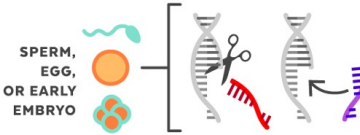
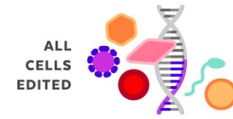
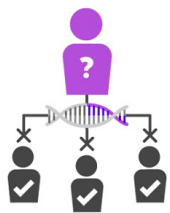
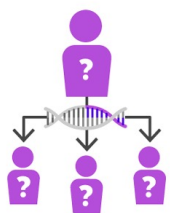
1. Derived from **Adenovirus**

- Will infect all cells even non-dividing ones
- The vectors will not be integrated in the genome, thus the transgene is diluted and eventually lost.

2. Derived from **Retrovirus**

- Many will infect only dividing cells, but others (HIV) can infect without host cell division.
- The transgene and the viral vector will be incorporated into the genome of the infected cells (more stable).

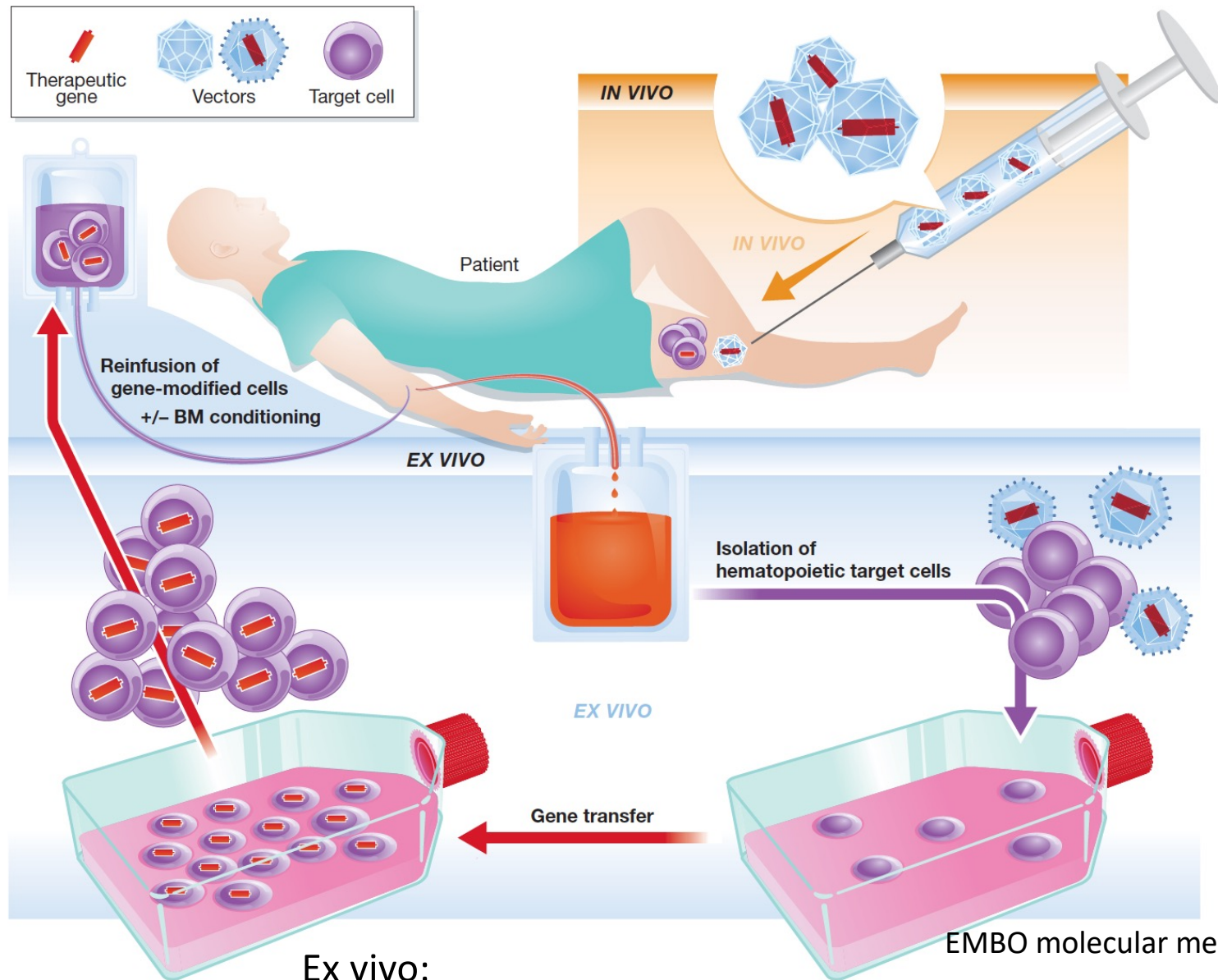
Two types of gene therapy:

	SOMATIC GENE EDITING	VS.	GERMLINE GENE EDITING
EDIT	 Somatic therapies target genes in specific types of cells (blood cells, for example).		 Germline modifications are made so early in development that any change is copied into all of the new cells.
COPY	 The edited gene is contained only in the target cell type. No other types of cells are affected.		 The edited gene is copied in every cell, including sperm or eggs.
RISKS	 Any changes, including potential off-target effects, are limited to the treated individual.		 If the person has children, the edited gene is passed on to future generations.
NEXT GENERATION	 The edited gene is not passed down to future generations.		
CONSENSUS	 Somatic cell therapies have been researched and tested for more than 20 years and are highly regulated.		 Human germline editing is new. Heritability of germline changes presents new legal and societal considerations.

Somatic cell: gene therapy refers to the transfer of a gene in somatic cells.

Germline: gene therapy refers to the transfer of a gene in all cells of an organism through the germline transmission

Concept of *in vivo* and *ex vivo* gene therapy



Ex vivo:

One can better control and monitor the efficiency of gene transfer .

ADA-SCID : Adenosine Deaminase - Severe Combined Immunodeficiency Disease.

ADA-SCID is a rare autosomal disease of the immune system (*Bubble boy disease*).

Affected individuals have essentially no immune system:

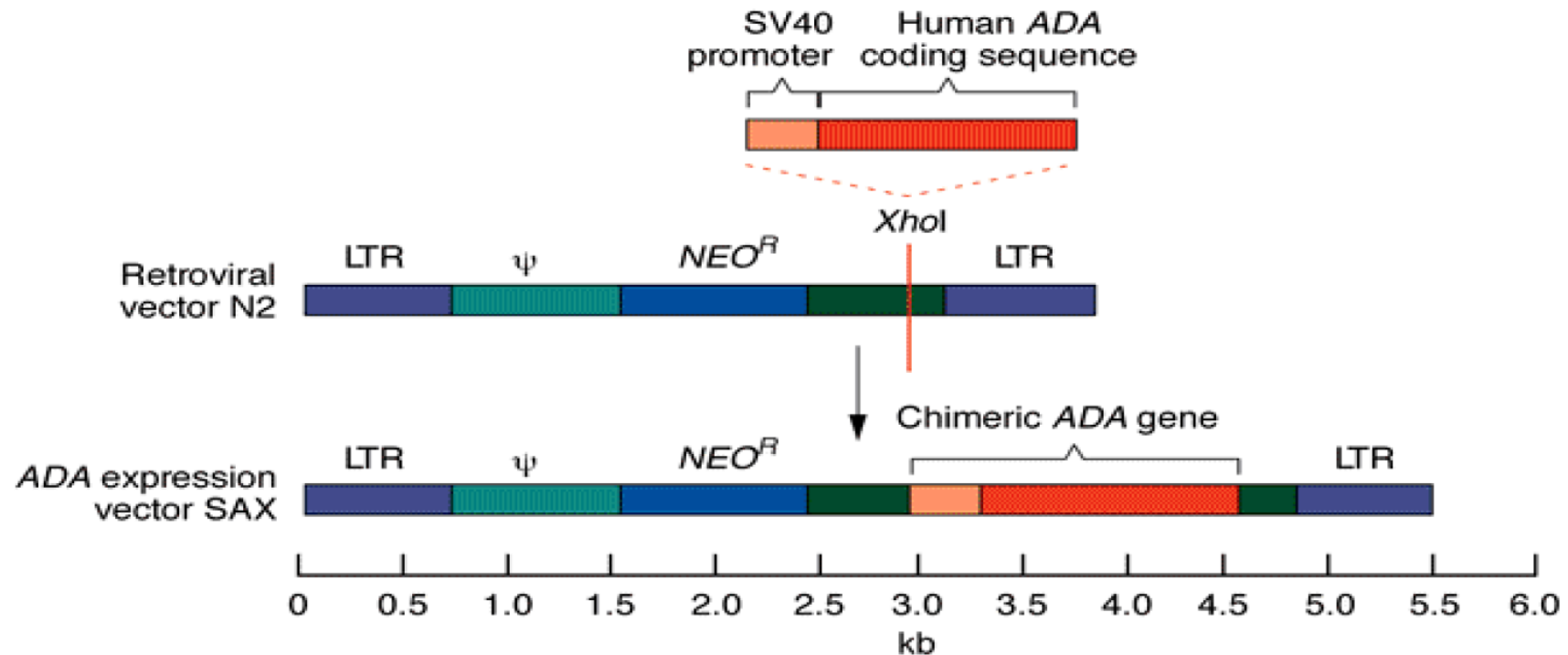
In absence of ADA, deoxyadenosine accumulates in T lymphocytes and eventually kill these cells (T lymphocytes are responsible for the stimulation of the antibody-producing B lymphocytes.)

First disease treated with gene therapy (1990).

The most common treatment for SCID is bone marrow transplantation

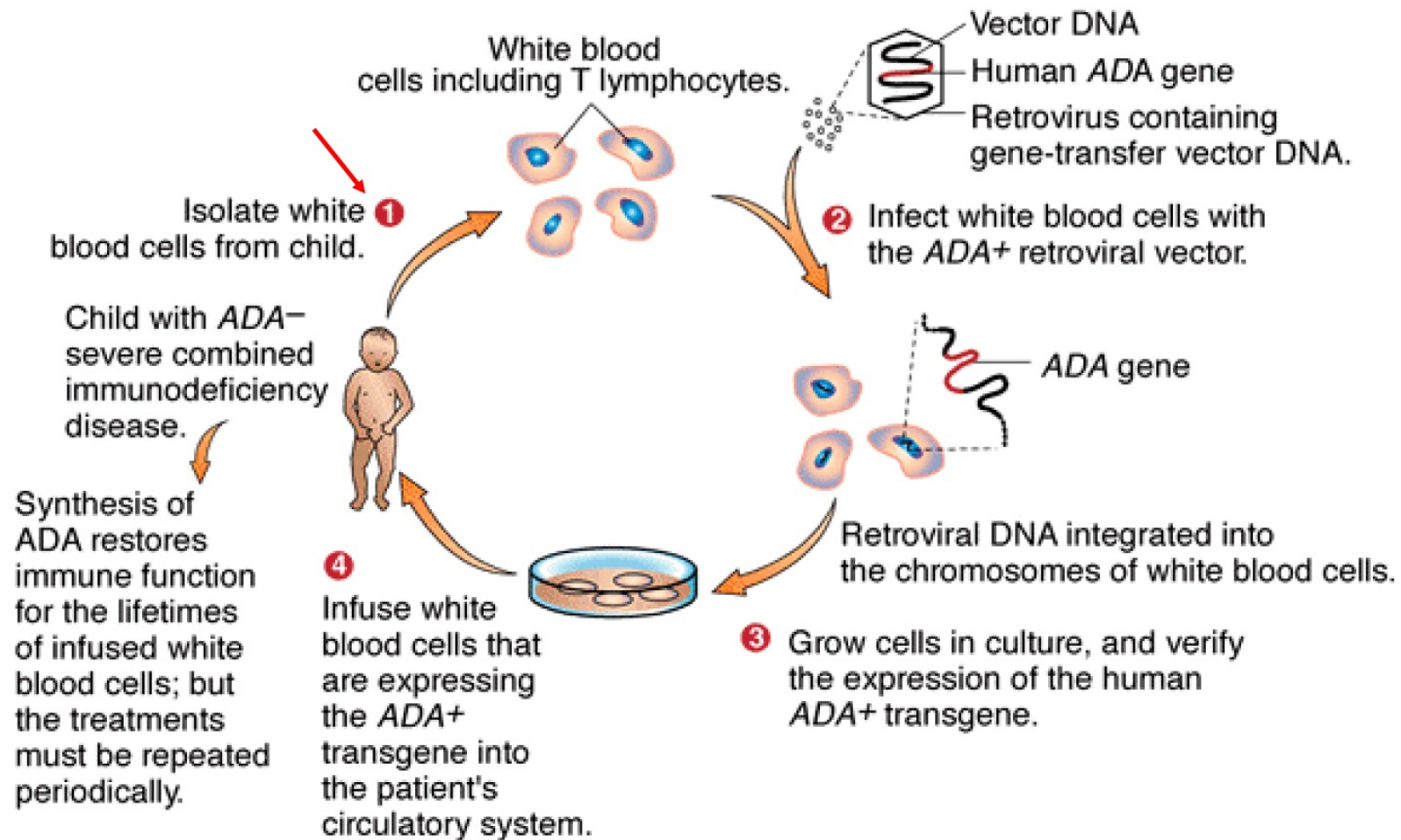


Construction of an expression vector



- The LTR are needed for the vector DNA integration in the cells.
- SV40 promoter will allow transgene expression in human cells.
- The NEO^R is needed for the selection of human cells that will produce the virus *in vitro*.

Partially curing the Severe Combined Immunodeficiency disease.



However.....

In 2007, four of the ten patients have developed leukemia due to the integration of the retroviral vector near a proto-oncogene.

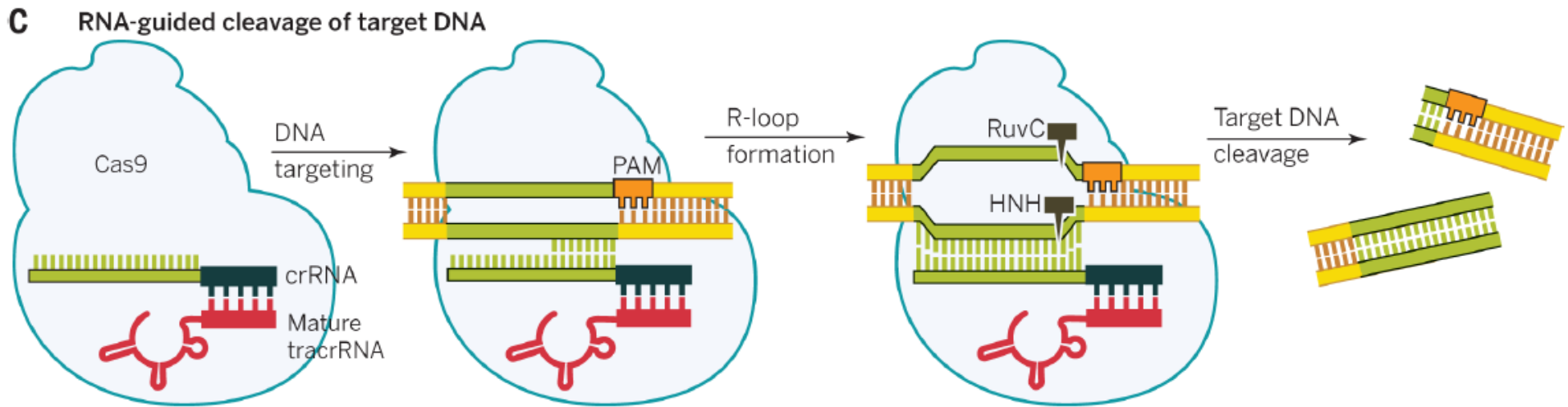
Issues with conventional gene therapy

- Integration sites cannot be controlled.
- Expression level of the rescue gene may not be optimal.
- Ex vivo experiment is limited to certain types of cells.

Potential solution: directly edit genomic sequences of stem cells from the patient

Tools to target specific sequences in the genome

The clustered regularly interspaced short palindromic repeats (CRISPR) system uses a RNA molecule to target specific DNA sequences.



To target different sequences in the genome, one has to simply change the crRNA sequences.

FDA NEWS RELEASE

FDA Approves First Gene Therapies to Treat Patients with Sickle Cell Disease

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For Immediate Release: December 08, 2023

Today, the U.S. Food and Drug Administration approved two milestone treatments, Casgevy and Lyfgenia, representing the first cell-based gene therapies for the treatment of sickle cell disease (SCD) in patients 12 years and older. Additionally, one of these therapies, Casgevy, is the first FDA-approved treatment to utilize a type of novel genome editing technology, signaling an innovative advancement in the field of gene therapy.

Induced Pluripotent stem cells (iPS)

Induction of Pluripotent Stem Cells from Mouse Embryonic and Adult Fibroblast Cultures by Defined Factors

Kazutoshi Takahashi¹ and Shinya Yamanaka^{1,2,*}

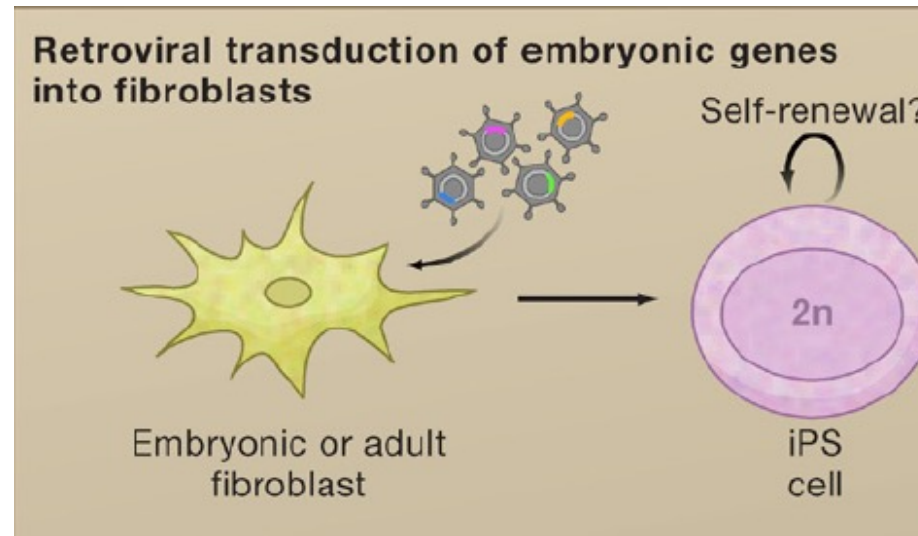
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DOI 10.1016/j.cell.2006.07.024

Cell. 2006



Overexpression of 4 proteins, Oct3/4, Sox2, c-Myc, and Klf4, in fibroblasts was sufficient to reprogram cells to become pluripotent stem cells.

2012 Nobel Prize for the discovery that mature cells can be reprogrammed to become pluripotent. (Unlike embryonic stem cells, there is no moral implication of using iPS)

Hopefully in the near future.....

